



Position A:

Acc:

$$a_A = \ddot{x}_A$$

vitesses $v_A = \dot{x}_A t + C$

avec $C = v$

donc

$$v_A = \dot{x}_A t + v$$

position $OA = \frac{1}{2} \ddot{x}_A t^2 + v t + C$

avec $C = d$

$$OA = \frac{1}{2} \ddot{x}_A t^2 + v t + d$$

Vitesse v :

Phase 1: $t \leq t_0$

acc:

$$a_B = 0$$

vitesses $v_B = v$

Position: $OB = vt$

$$x_B(t_0) = v t_0$$

Phase: a $t > t_0$

accel:

$$a_B = x_B$$

when: $x_B = x_B t + C$

a $t = t_0$

$$v_B(t_0) = x_B t_0 + C = v$$

$$C = v - x_B t_0$$

done

$$v_B = x_B t + v - x_B t_0$$

$$v_B = x_B (t - t_0) + v$$

Position:

$$O_B = \frac{1}{2} x_B t^2 + v t + C$$

a $t = t_0$

$$-\frac{1}{2} x_B t_0^2 + v t_0 + C = v t_0$$

$$C = \frac{1}{2} x_B t_0^2$$

$$O_B = \frac{1}{2} x_B t^2 + (v - \frac{1}{2} x_B t_0^2) + \frac{1}{2} x_B t_0^2$$

$$= \frac{1}{2} x_B (t - t_0)^2 + v t$$

2.

on cherche quand

$$OB = OA$$

$$\frac{1}{2} \ddot{x}_A t^2 + vt + d = \frac{1}{2} \ddot{x}_B (t - t_0)^2 + vt$$

$$\frac{1}{2} \ddot{x}_A t^2 + vt + d - \frac{1}{2} \ddot{x}_B (t - t_0)^2 + vt = 0$$

A.N à la calculatrice

$$t = 11,0 \text{ s}$$

$$\bullet OA(11) = 289 \text{ m}$$